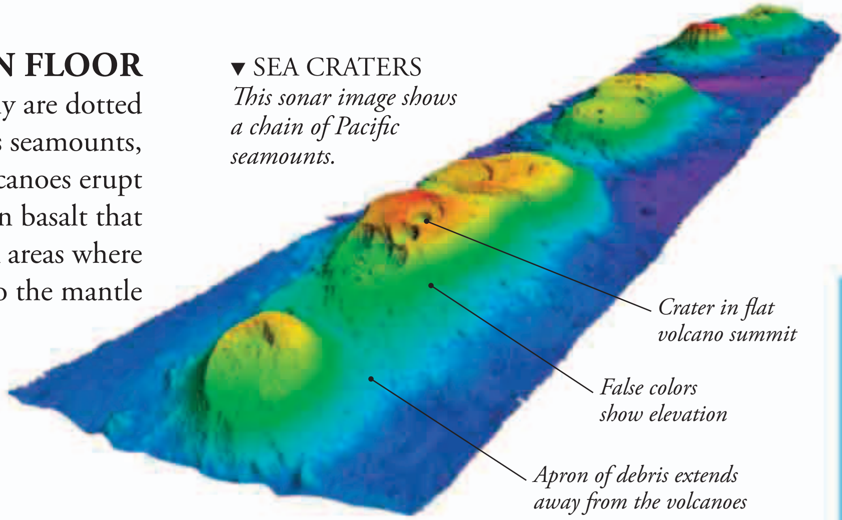
OCEAN FLOOR

The ocean floors are far from featureless. Many are dotted with submerged extinct volcanoes known as seamounts, which sometimes form long chains. Other volcanoes erupt from mid-ocean ridges, producing the molten basalt that forms the ocean floors. Deep trenches mark areas where old oceanic crust is being drawn into the mantle

▼ SEA CRATERS
This sonar image shows a chain of Pacific seamounts.

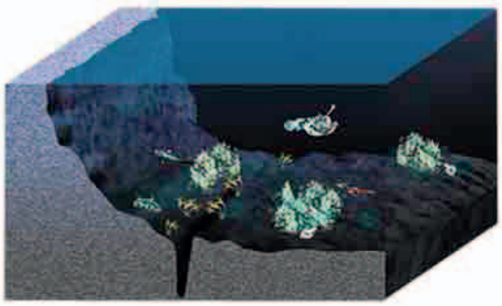


LIGHT AND HEAT

Scientists divide the ocean into vertical zones, depending on how much sunlight they receive. Near the surface is the sunlit zone, which is bright enough for photosynthesis to take place. Most ocean life is found here. Below 660 ft (200 m) is the twilight zone, where there is only blue light and the water is colder. Many of the animals here rise to the surface at night to feed on plankton. Below 3,300 ft (1,000 m) is the dark zone, where there is no sunlight and the water is close to freezing. Most animals here feed on dead organisms that have drifted down from above.

Sunlit zone

Twilight zone



Dark zone

EXPLORING THE DEEP

Most of what we know about the deep ocean floors has been discovered through remote sensing—using devices at the surface to detect what lies beneath. Scientists can also use submersibles—underwater craft, such as the one above—but the difficulty of reaching the ocean floors means that only a tiny fraction has been explored.

SUBMARINE CANYONS

On average, the oceans are almost 2.5 miles (4 km) deep, but their fringes are much shallower. This is because the seabed near the shore is the submerged edge of a continent called a continental shelf. Rivers flowing into the sea can cut deep canyons in the sea beds of the continental shelves.